mitochondria but in animals 38 ATP molecules are produced per glucose molecule. During anaerobic respiration only 2 ATP molecules are produced, therefore anaerobic respiration is less efficient than aerobic respiration. The respiratory quotient (RQ) is the ratio of carbon dioxide production to oxygen consumption and reflects the relative contributions of fat, carbohydrate, and protein to the oxidation. Pentose phosphate pathway is an alternative pathway to glycolysis and TCA cycle for oxidation of glucose. It occurs in cytoplasm of both prokaryotes and eukaryotes.

Evaluation

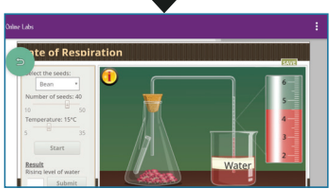
1. The number of ATP molecules formed by complete oxidation of one molecule of pyruvic acid is
   a. 12 b. 13 c. 14 d. 15
2. During oxidation of two molecules of cytosolic NADH + H^+, number of ATP molecules produced in plants are
   a. 3 b. 4 c. 6 d. 8
3. The compound which links glycolysis and Krebs cycle is
   a. succinic acid b. pyruvic acid
   c. acetyl CoA d. citric acid
4. Assertion (A): Oxidative phosphorylation takes place during the electron transport chain in mitochondria.
   Reason (R): Succinyl CoA is phosphorylated into succinic acid by substrate phosphorylation.
   a. A and R is correct. R is correct explanation of A
   b. A and R is correct but R is not the correct explanation of A
   c. A is correct but R is wrong
   d. A and R is wrong.
5. Which of the following reaction is not involved in Krebs cycle.
   a. Shifting of phosphate from 3C to 2C
   b. Splitting of Fructose 1,6 bisphosphate of into two molecules 3C compounds.
   c. Dephosphorylation from the substrates
   d. All of these
6. What are enzymes involved in phosphorylation and dephosphorylation reactions in EMP pathway?
7. Respiratory quotient is zero in succulent plants. Why?
8. Explain the reactions taking place in mitochondrial inner membrane.
9. What is the name of alternate way of glucose breakdown? Explain the process involved in it?
10. How will you calculate net products of one sucrose molecule upon complete oxidation during aerobic respiration as per recent view?

CR2HJX

ICT Corner

Rate of Respiration

Let's estimate **rate of respiration**



Steps

- Scan the QR code or go to google play store
- Type online labs and install it.
- Select biology and select rate of respiration
- Click theory to know the basic about respiration
- Register yourself with mail-id and create password to access online lab simulations

Activity

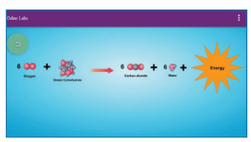
- Press simulation to do the rate of respiration.
- Conclude your observations.


Step 1



Step 2


Step 3


Step 4

B166_11_BOT_EM

URL: https://play.google.com/store/apps/details?id=in.edu.olabs.olabs&hl=en

Alternate web: http://www.sumanasinc.com/webcontent/animations/content/cellularrespiration.html

** Pictures are indicative only*

XIth_BIO-BOTANY_Ch_14-EM.indd 275 5/17/2020 11:29:40 PM